UPSA
UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA
Scholarship with Professionalism

NETWORK AND SYSTEMS ADMINISTRATION

WEEK 5: USER MANAGEMENT AND SYSTEMS SECURITY

Users • Groups • Authentication • Permissions • Least Privilege • Hardening • Logging

Practical Environment: Kali Linux VM

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WEEK 5 ROADMAP

From identity to secure administration

1. **Users & Groups** — Who needs access?
2. **Authentication** — How do we prove identity?
3. **Permissions** — What may each user do?
4. **Delegation** — Who may administer?
5. **Hardening** — How do we reduce risk?
6. **Evidence** — How do we prove it?

INDIVIDUAL PRACTICAL

Create and manage test users/groups, apply permissions, and inspect selected security controls on the Linux VM.

GROUP ACTIVITY

Convert organisational roles into an access-control matrix, implement sample accounts, test access and conduct a hardening review.

LEARNING OUTCOMES

By the end of Week 5, students should be able to:

- Explain the purpose of user accounts, groups, authentication and authorisation in Linux administration.
- Create and manage test users and groups safely in a controlled lab environment.
- Interpret ownership and permission information from Linux command output.
- Apply basic file permissions and ownership changes using appropriate commands.
- Explain least privilege and administrative delegation using sudo.
- Inspect selected security controls, including failed services, open sockets, firewall status and warning logs.
- Translate organisational roles into an access-control matrix and test authorised and denied access.
- Document unnecessary exposure or excessive privileges and recommend corrective action.

WHY WEEK 5 MATTERS

Infrastructure becomes risky when access is uncontrolled.

FROM WEEK 1	FROM WEEKS 2–4	NOW WEEK 5
We identified infrastructure, documentation, change control and administrator responsibility.	We inspected computing resources, services, workstations and network configuration.	We control who can use those resources, what they can do, and how security evidence is checked.

A well-connected system is not necessarily a safe system. Access must be managed deliberately.

USER MANAGEMENT AND SYSTEMS SECURITY

Two sides of the same administrative responsibility.

USER MANAGEMENT

Creating, organising, modifying, disabling and documenting identities so people and services can do legitimate work.

SYSTEMS SECURITY

Protecting systems through authentication, permissions, hardening, patching, logging and monitoring.

ADMINISTRATIVE CONTROL

Ensuring privileged actions are authorised, traceable and limited to what the task requires.

The goal is not to stop work; the goal is to make authorised work safe, reliable and accountable.

THE ACCESS CONTROL STACK

A practical model for Week 5

USER OR SERVICE — Who is requesting access?

AUTHENTICATION — How is identity verified?

GROUPS / ROLES — Which category of access applies?

PERMISSIONS / OWNERSHIP — What object and operation are allowed?

PRIVILEGED DELEGATION — Is elevated authority justified?

LOGGING / REVIEW — Can actions and risks be checked?

USERS: HUMAN, SERVICE AND ADMINISTRATIVE ACCOUNTS

Every account should have a purpose, an owner and an access boundary.

- Human accounts represent individual people and should support accountability.
- Service accounts support applications or background services and should have tightly limited rights.
- Administrative accounts or privileges should be controlled and used only for authorised tasks.
- Test accounts in this class must remain within the Linux VM and should not reuse personal passwords.
- Disabled or abandoned accounts create unnecessary exposure and should be reviewed.

ADMINISTRATOR'S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

GROUPS: MANAGING ACCESS AT SCALE

Groups allow administrators to assign permissions to roles rather than to individuals one by one.

- A group represents a shared access need, for example accounts, sales, lecturers, students or support.
- Users may belong to a primary group and one or more supplementary groups.
- Group-based access is easier to audit than many individual exceptions.
- Poorly named or unmanaged groups become difficult to interpret later.
- Group membership should follow organisational role and least-privilege logic.

ADMINISTRATOR'S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

AUTHENTICATION AND PASSWORDS

Authentication proves identity before authorisation decisions are made.

- Authentication answers: “Are you the identity you claim to be?”
- Passwords remain common but must be protected, changed responsibly and never shared.
- Administrators should understand where account information is stored or obtained.
- Password resets are administrative actions and should follow verification procedures.
- Multi-factor authentication may be required in stronger environments, although today focuses on local Linux practice.

ADMINISTRATOR’S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

AUTHORISATION, PERMISSIONS AND OWNERSHIP

Authorisation answers what an authenticated identity may do.

- Linux permission decisions depend on user identity, group membership, ownership and mode bits.
- Ownership identifies the user and group associated with a file or directory.
- Read, write and execute permissions mean different things for files and directories.
- Permission changes should support a clear business or lab requirement.
- Administrators should test both allowed and denied access.

ADMINISTRATOR'S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

LEAST PRIVILEGE

Give only the access required for the assigned task, and no more.

- Least privilege reduces accidental damage and limits the effect of compromised accounts.
- Users should not receive administrative rights simply for convenience.
- Groups should not be granted broad access when narrow access is sufficient.
- Temporary access should be removed when the task is complete.
- Evidence of least privilege includes clear membership, permissions and access-test results.

ADMINISTRATOR'S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

ADMINISTRATIVE DELEGATION

Delegation gives limited administrative authority without uncontrolled root access.

- sudo can allow selected users to run selected commands with elevated privileges.
- Delegation should match a role, not personal favour or convenience.
- Administrators must understand what their own sudo rights allow.
- Privileged commands should be traceable and used deliberately.
- Excessive delegation is a security weakness, even when no attack has yet occurred.

ADMINISTRATOR'S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

SYSTEM HARDENING

Hardening reduces unnecessary exposure before failure or attack occurs.

- Disable or remove unnecessary accounts, services and open network listeners.
- Apply security updates and review failed or misconfigured services.
- Use firewall rules to reduce reachable services where appropriate.
- Protect sensitive files and administrative credentials.
- Document hardening decisions so that they can be reviewed or reversed safely.

ADMINISTRATOR'S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

PATCHING AND UPDATE DISCIPLINE

Patching is security work, operational work and change-control work.

- Updates correct defects, vulnerabilities and compatibility problems.
- Patch timing should consider risk, service availability and testing requirements.
- Systems that cannot be patched immediately may require compensating controls.
- Patch decisions should be documented, especially for critical systems.
- After patching, verify service health and log for errors.

ADMINISTRATOR'S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

FIREWALL CONFIGURATION AND EXPOSURE

A service that is running may or may not need to be reachable from the network.

- Firewalls help control which network traffic is allowed to reach the host.
- Open ports should match legitimate services and organisational need.
- The ss command shows listening sockets; ufw status shows firewall posture on systems using UFW.
- Firewall changes can block legitimate users if poorly planned.
- Document firewall rules and test expected access after changes.

ADMINISTRATOR'S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

AUDITING, LOGGING AND REVIEW

Security controls are weak when no one checks the evidence.

- Logs record events that may reveal misconfiguration, failure or suspicious activity.
- journalctl can filter systemd journal messages by priority and time.
- Audit thinking asks: who did what, when, from where and with what outcome?
- Logs may contain sensitive data and must be handled responsibly.
- Warnings are not always emergencies, but repeated warnings should be investigated.

ADMINISTRATOR'S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

SECURE ADMINISTRATIVE PRACTICE

Good administrators protect the system and protect the trust placed in them.

- Use privileged commands only when required and authorised.
- Never share administrative passwords or reuse weak lab passwords in real environments.
- Record significant changes and preserve enough evidence for explanation.
- Communicate planned changes and service interruptions clearly.
- Verify outcomes rather than assuming a command succeeded.

ADMINISTRATOR'S QUESTION

What access is needed, who approved it, and how will we verify that it is safe?

EVIDENCE

Command output, access test, group membership, permission listing, log entry or change record.

INDIVIDUAL PRACTICAL WORKFLOW

Create, control, inspect and document in the Linux VM.

IDENTIFY

whoami
id

DISCOVER

getent passwd
getent group

CREATE

sudo useradd -m USER
sudo passwd USER

ORGANISE

sudo groupadd GROUP
sudo usermod -aG GROUP USER

CONTROL

ls -l
chmod
chown

INSPECT SECURITY

sudo -l
systemctl --failed
ss -tulpn
sudo ufw status verbose
sudo journalctl -p warning --since today

COMMAND: whoami

Interpret the output, do not only capture it.

$ whoami

WHAT IT TELLS YOU

Shows the effective username for the current shell session. It answers the immediate question: “Which account am I operating as right now?”

WHAT TO RECORD

Record the username returned. Explain why knowing the current identity matters before running administrative commands.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: id

Interpret the output, do not only capture it.

$ id

WHAT IT TELLS YOU	WHAT TO RECORD
Displays user ID, primary group ID and supplementary group memberships for the current or specified user. It links the account to access decisions.	Record UID, GID and groups. Interpret whether the user belongs to any privileged or special group.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: getent passwd

Interpret the output, do not only capture it.

$ getent passwd

WHAT IT TELLS YOU

Queries the configured name-service sources for user account entries. It is safer than manually opening local files because it respects the system's configured account sources.

WHAT TO RECORD

Identify selected account fields such as username, UID, GID, home directory and shell. Do not expose sensitive information unnecessarily.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: getent group

Interpret the output, do not only capture it.

$ getent group

WHAT IT TELLS YOU	WHAT TO RECORD
Queries group entries and shows group names, group IDs and listed members. It helps administrators review role-based access structures.	Record relevant groups and members. Explain why group membership can change access without directly changing file permissions.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: sudo useradd -m USER

Interpret the output, do not only capture it.

$ sudo useradd -m testuser

WHAT IT TELLS YOU	WHAT TO RECORD
Creates a new test user and, with -m, creates a home directory. This demonstrates account provisioning in a controlled lab.	Use a safe test name. Verify creation using getent passwd and check the home directory where appropriate.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: sudo passwd USER

Interpret the output, do not only capture it.

$ sudo passwd testuser

WHAT IT TELLS YOU	WHAT TO RECORD
Sets or changes a password for a user. This is an administrative action and should be handled with care.	Use a temporary lab password only. Never display or record real passwords in evidence.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: sudo groupadd GROUP

Interpret the output, do not only capture it.

$ sudo groupadd accounts

WHAT IT TELLS YOU	WHAT TO RECORD
Creates a new group that can be used to organise role-based access.	Choose a meaningful group name. Verify creation using getent group.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: sudo usermod -aG GROUP USER

Interpret the output, do not only capture it.

$ sudo usermod -aG accounts testuser

WHAT IT TELLS YOU

Adds a user to a supplementary group. The -a option appends; without it, existing supplementary memberships may be replaced.

WHAT TO RECORD

Verify membership using id USER or getent group. Explain why the append option matters.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: ls -l

Interpret the output, do not only capture it.

$ ls -l

WHAT IT TELLS YOU	WHAT TO RECORD
Lists files with ownership and permission details. It is the main evidence command before and after changing permissions or ownership.	Identify file type, owner, group and rwx permissions for user, group and others.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: chmod

Interpret the output, do not only capture it.

$ chmod 640 FILE

WHAT IT TELLS YOU

Changes file permission bits. The administrator must know what access is being granted or removed before applying it.

WHAT TO RECORD

Explain the permission change in words, then verify using ls -l. Test authorised and denied access where possible.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: chown

Interpret the output, do not only capture it.

$ sudo chown USER:GROUP FILE

WHAT IT TELLS YOU	WHAT TO RECORD
Changes file owner and/or group. Ownership affects which permission category applies to an access request.	Use carefully because ownership changes can break applications or expose data. Verify with ls -l.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: sudo -l

Interpret the output, do not only capture it.

$ sudo -l

WHAT IT TELLS YOU	WHAT TO RECORD
Lists the commands the current user is allowed to run through sudo. It helps reveal administrative delegation.	Record whether sudo access exists and what it permits. Discuss whether the permission is narrow, broad or excessive.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: systemctl --failed --no-pager

Interpret the output, do not only capture it.

$ systemctl --failed --no-pager

WHAT IT TELLS YOU

Shows failed systemd units. Failed services may indicate misconfiguration, failed updates or security-relevant instability.

WHAT TO RECORD

Record failed units and interpret whether any require investigation. Do not ignore repeated service failures.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: ss -tulpn

Interpret the output, do not only capture it.

$ ss -tulpn

WHAT IT TELLS YOU

Shows listening TCP/UDP sockets and associated processes where permission allows. It helps identify exposed services.

WHAT TO RECORD

Compare listening services with what the host is expected to provide. Open listeners should have a legitimate purpose.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: sudo ufw status verbose

Interpret the output, do not only capture it.

$ sudo ufw status verbose

WHAT IT TELLS YOU

Shows UFW firewall status and rule posture on systems where UFW is installed and active.

WHAT TO RECORD

Record whether the firewall is active and what rules or defaults are shown. Explain that absence of UFW does not mean absence of all firewalling.

Evidence standard: command + relevant output + interpretation + administrative meaning.

COMMAND: sudo journalctl -p warning --since today

Interpret the output, do not only capture it.

$ sudo journalctl -p warning --since today

WHAT IT TELLS YOU

Filters journal messages to warnings and more serious entries since today. It supports quick security and stability review.

WHAT TO RECORD

Identify repeated warnings, authentication-related messages or service errors. Do not publish sensitive log details.

Evidence standard: command + relevant output + interpretation + administrative meaning.

PRACTICAL EVIDENCE CHECKLIST

Each student should leave with a clear security administration record.

Area	Command evidence	Interpretation required
Identity	whoami, id	Current user, UID/GID and group membership
Account sources	getent passwd, getent group	Relevant user/group records and role meaning
Provisioning	useradd, passwd	New test user created and password set safely
Group management	groupadd, usermod -aG	Role group created and membership verified
Permissions	ls -l, chmod, chown	Before/after permission and ownership evidence
Delegation	sudo -l	Scope of administrative authority
Security review	systemctl, ss, ufw, journalctl	Failed units, exposed services, firewall posture and warnings

GROUP ACTIVITY: ACCESS-CONTROL MATRIX

Translate organisational roles into controlled access.

- Use the same small Ghanaian organisation selected in earlier weeks, or choose a new small scenario approved by the lecturer.
- Identify at least four organisational roles, for example manager, cashier, accountant, technician or guest.
- Identify at least four resources, for example payroll file, sales folder, records system, printer or backup directory.
- Decide read, write, execute/admin or no access for each role-resource pair.
- Justify one access decision using least privilege.
- Identify one excessive privilege and propose the correction.

Role / Resource	Sales	Payroll	Backup
Manager	R	R	No
Accountant	R	RW	No
Cashier	RW	No	No
Technician	No	No	Admin

GROUP IMPLEMENTATION: ACCOUNTS, GROUPS AND PERMISSIONS

Turn selected matrix decisions into Linux evidence.

- Create representative test users for two or three roles.
- Create role groups that match the access-control matrix.
- Create a test folder or file that represents an organisational resource.
- Assign ownership and permissions according to the group decision.
- Test an authorised access case and a denied access case.
- Record the command evidence and explain what worked or failed.

GOOD TEST

A good test proves both what is allowed and what is denied.

GROUP OUTPUT

Matrix + commands + evidence + corrective action.

GROUP HARDENING REVIEW

Find unnecessary exposure before it becomes an incident.

Accounts	Groups	Permissions	Privileges
Unused or unnecessary users?	Excessive group membership?	Files writable by too many users?	Too much sudo access?

Services	Network	Firewall	Logs
Failed or unnecessary services?	Unexpected listening sockets?	Firewall disabled or overly permissive?	Repeated warnings or suspicious messages?

Document excessive privileges or unnecessary exposure and state the corrective action taken.

GROUP PRESENTATION GUIDE

Five minutes per group: evidence, not storytelling alone.

- 1 minute: organisation, roles and resources.
- 1 minute: access-control matrix and least-privilege justification.
- 1 minute: Linux implementation evidence.
- 1 minute: authorised and denied access test evidence.
- 1 minute: hardening review finding and corrective action.
- Peer question: What access or exposure did the group overlook?

LECTURER CHECK

Matrix present? Commands shown? Tests verified? Risk explained?

EVIDENCE RULE

No interpretation, no full credit.

WEEK 5 KEY TAKEAWAYS

Secure administration is deliberate, documented and tested.

1. User accounts represent identities and must be controlled throughout their lifecycle.
2. Groups help scale access management, but unmanaged groups create risk.
3. Permissions and ownership translate policy into enforceable Linux controls.
4. Least privilege and sudo delegation reduce unnecessary administrative exposure.
5. Hardening includes patching, firewall posture, service review, logging and secure habits.
6. Good evidence proves both authorised access and denied access.